DevOps Task Manager Application using React and Node.js

A course project report submitted in partial fulfilment of the requirement of

DEVOPS AND FULLSTACK

by

J.Sujith (2303A51327)

Under the guidance of

Dr. N.VENKATESH

Asst. Professor, Department of CS & AI

SR University, Warangal

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ABSTRACT

- This project, *“Explainable AI for Kidney Transplant Prediction,”* focuses on developing a clinical decision-support system using machine learning and explainable AI techniques. The system predicts kidney transplant outcomes such as graft survival and complication risk using models like Random Forest, Gradient Boosting, and Support Vector Regression (SVR).
- A synthetic dataset of 1000 patient records was generated to simulate real-world transplant scenarios. Data preprocessing techniques such as One-Hot Encoding and Standardization were applied, followed by feature engineering and Recursive Feature Elimination (RFE) to improve model performance.
- To enhance transparency and trust in predictions, SHAP (SHapley Additive exPlanations) was used to interpret model outputs and identify important features influencing decisions. The system is integrated into a web-based dashboard built using HTML, CSS, and JavaScript for user interaction.
- The results demonstrate that Random Forest with SHAP achieved the best performance with Mean Squared Error (MSE) values of approximately 0.089 for graft survival and 0.085 for complication risk. This project highlights the importance of explainable AI in healthcare for better decision-making and improved patient outcomes.

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CHAPTER 1: INTRODUCTION

1.1 About the Project

The project aims to develop an intelligent and explainable system for predicting kidney transplant outcomes. In the healthcare domain, accurate prediction of graft survival and complication risks is critical for improving patient care and clinical decision-making.

This system combines machine learning models with explainable AI techniques to provide both accurate predictions and transparent insights. A web-based dashboard is designed to allow healthcare professionals to interact with the system easily, visualize results, and understand model decisions.

1.2 Objectives

- To build a machine learning model for kidney transplant prediction
- To compare multiple algorithms like Random Forest, Gradient Boosting, and SVR
- To improve model performance using feature engineering and selection techniques
- To implement explainable AI using SHAP for interpretability
- To develop a user-friendly web dashboard for visualization and interaction

CHAPTER 2: LITERATURE REVIEW

2.1 DevOps and Full-Stack Application Development

Modern healthcare applications require integration of machine learning with full-stack development. Full-stack development involves building both frontend (user interface) and backend (data processing and ML models).

DevOps practices help in automating deployment, testing, and integration of applications, ensuring faster development cycles and reliability.

In this project:

- Frontend is developed using HTML, CSS, and JavaScript
- Backend ML pipeline is implemented using Python and scikit-learn

- Visualization tools like Chart.js are used for displaying results

This integration ensures a smooth workflow from data processing to user interaction.

CHAPTER 3: METHODOLOGY

3.1 System Design

The system consists of two main components:

1. **Frontend Dashboard** – for user interaction and visualization
2. **ML Backend Pipeline** – for prediction and analysis

The user logs into the system, inputs data, and receives predictions along with explanations.

3.2 Working Process

1. Data is generated (synthetic dataset with 1000 samples)
2. Data preprocessing is applied (encoding and scaling)
3. Features are engineered and selected
4. Models are trained using training data
5. Predictions are made on test data
6. SHAP is used to explain model predictions
7. Results are displayed on the dashboard

3.3 Technologies Used

- HTML5, CSS3, JavaScript
- Python
- Machine Learning algorithms (RF, GB, SVR)
- SHAP for explainability

3.4 Tools and Technologies Used

- VS Code (development environment)

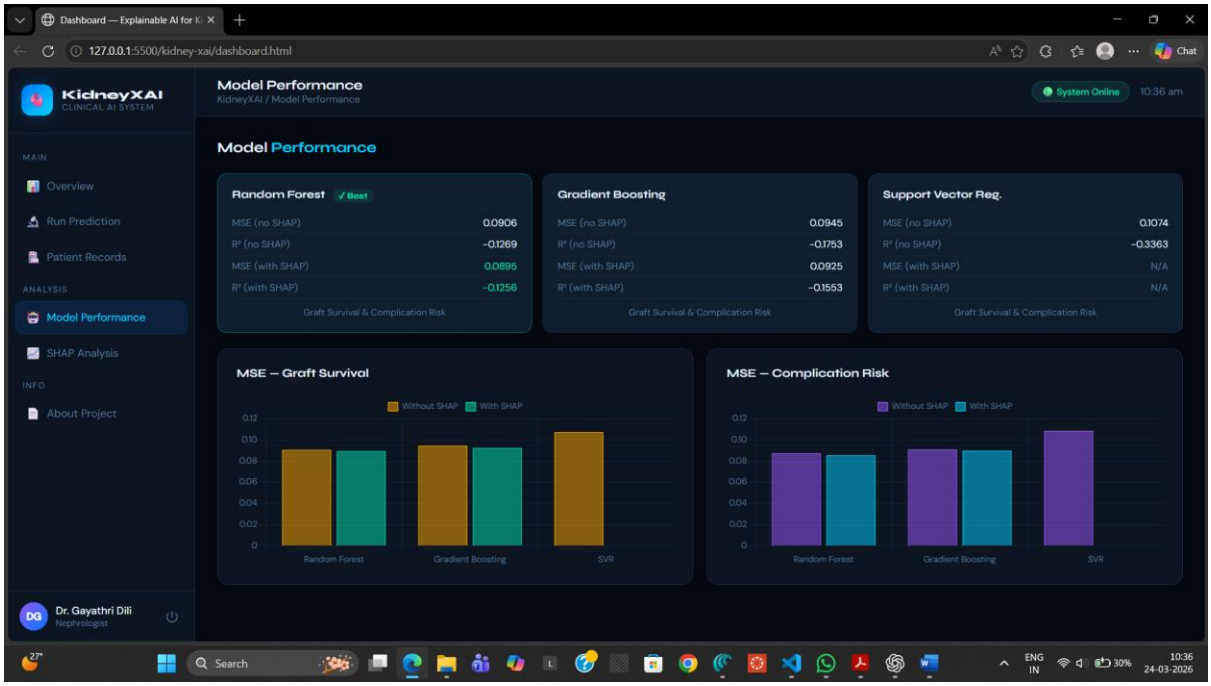
- Live Server (for running frontend)
- Python libraries: pandas, NumPy, scikit-learn, Matplotlib
- Chart.js (for visualization)
- GitHub (version control and deployment)

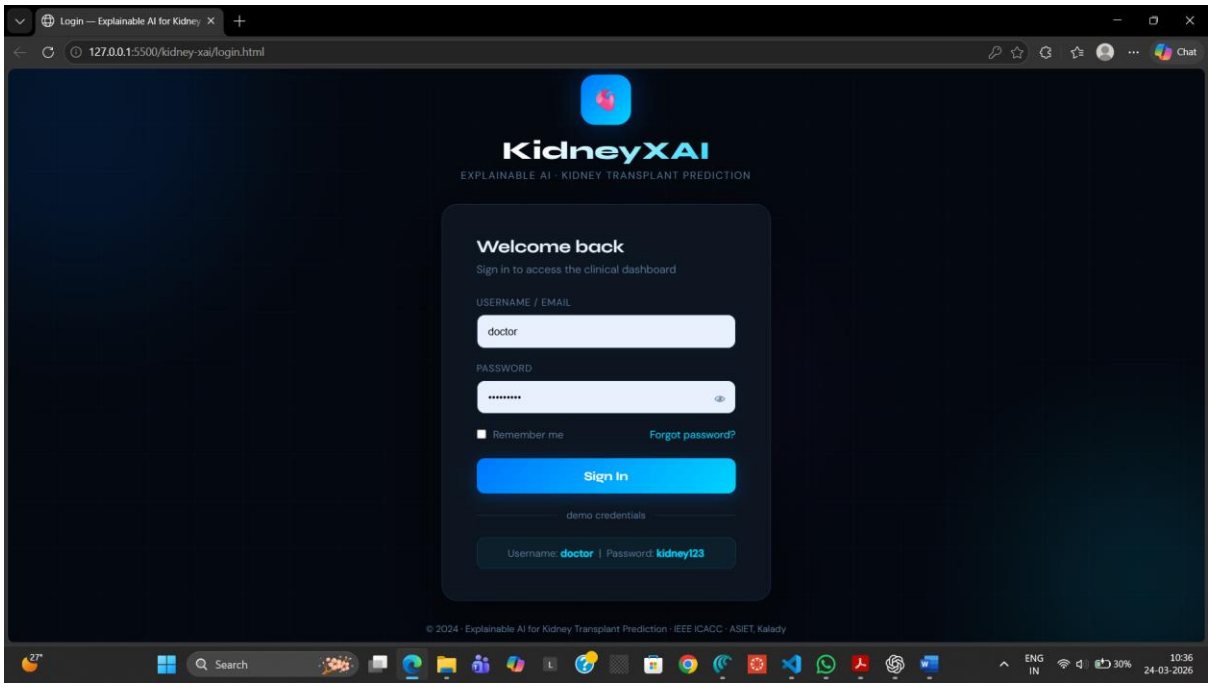
CHAPTER 4: RESULTS AND DISCUSSION

4.1 Results

The performance of models is evaluated using Mean Squared Error (MSE):

- Random Forest (Best):
  - Graft Survival: ~0.089
  - Complication Risk: ~0.085
- Gradient Boosting: Slightly higher error
- SVR: Lowest performance among the models





4.2 Analysis

- Random Forest performed best due to its ability to handle complex relationships
- Feature selection improved model accuracy
- SHAP provided clear insights into feature importance

4.3 Discussion

The integration of explainable AI enhances trust in predictions, which is crucial in healthcare applications. The web dashboard makes the system accessible and easy to use for medical professionals.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Summary

This project successfully developed an explainable AI-based system for kidney transplant prediction using machine learning and a web interface.

5.2 Conclusion

The results show that combining machine learning with explainability improves both accuracy and transparency. Random Forest with SHAP proved to be the most effective approach.

5.3 Future Scope

- Use real-world clinical datasets
- Deploy the model as a cloud-based application
- Integrate with hospital management systems
- Improve model performance using deep learning
- Add real-time prediction features